

Unintended consequences of the new regulatory paradigm: Real estate prices, and financing the economy

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1. Summary of the research plan

Since the 2007 financial crisis, banking sector regulation has significantly expanded (e.g., Baker and Wurgler, 2015). Existing academic literature suggests that these regulations enhance financial system stability (e.g., Gauthier et al., 2012), but also introduce potential adverse effects, such as credit rationing and altered risk-taking incentives (e.g., Behn et al., 2016; Duffie, 2018). Our project will explore three unexamined adverse consequences resulting from recent regulatory shifts, of relevance to both practitioners and policymakers.

The first subproject investigates the repercussions of abolishing the advanced internal ratings-based approach (AIRB) for the large corporate segment (revenues of more than EUR 500 million).¹ This change necessitates a uniform 40% loss given default (LGD) rate for all senior unsecured loans, potentially leading to cross-subsidization from low LGD firms to high LGD firms. This, in turn, may encourage low LGD borrowers to seek alternative financing methods like equity or public debt, possibly triggering a gradual decline in bank lending to this segment. We intend to identify the effects within a regression discontinuity design.

The second subproject investigates the impact of leverage regulation on real estate prices. Given the remarkable upsurge in real estate values and outstanding mortgages (Richter and Ziegler, 2021), concerns regarding the existence of a housing bubble have emerged (Glaeser, 2013). Our research is dedicated to uncovering the effects of leverage regulation on real estate market dynamics. We analyze a change in the self-regulation practices of Swiss banks that restrict eligible equity in their maximum loan-to-value ratio calculations for borrowers. Using the particularities of the rule adjustments our research can shed light on the mechanisms connecting leverage to real estate prices, local price variations, and market segmentation that have remained unexplored in the previous literature. We use a difference-in-difference approach for identification.

The third subproject concentrates on bank regulation and loan maturity. Banks employ a regulator-defined formula that includes a maturity adjustment (BCBS, 2017a) for calculating capital requirements related to loans. This adjustment increases capital requirements with (legal) loan maturity, raising long-term loan costs. Consequently, companies opt for shorter loan durations, potentially increasing refinancing and interest rate risk on corporate balance sheets. Our novel approach utilizes the granular instrumental variable (GIV) method for estimating causal supply elasticities (Gabaix and Koijen, 2020), a method not previously employed in the literature in this context.

¹ The Basel regulation offers banks the choice between two approaches to determine their risk weighted assets (RWA). With the standardized approach, the regulatory framework determines the RWA. With the internal ratings-based approach (IRB) the banks model certain risk parameters themselves and determine the RWA based on a formula given by the regulator. Within the IRB approach banks can choose between AIRB, where banks model the probability of default (PD), Loss given default (LGD) and exposure at default (EAD) or the foundation IRB approach (FIRB), where they only model the PD.

2. Research plan

2.1. Current state of research in the field

Bank Regulation and corporate financing decisions

The first subproject titled “Bank Regulation and corporate financing decisions” is concerned with the effect of bank regulation on corporate financing decisions. The literature on the impact of bank regulation has mainly been concerned with the questions how regulation impacts banks (e.g., Gropp and Heider, 2010, Celerier et al., 2020) and whether regulation makes banks safer (e.g., Gauthier et al., 2012). Moreover, regulations effects on the availability of credit, economic development and the rise of shadow banking (e.g., Kovner and van Tassel, 2022; Glancy, 2018; Santos and Winton, 2019; Fraisse et al., 2020 and Baker and Wurgler, 2015 and Irani et al., 2021) have been studied.

While there is an extensive body of research that examines the impact of regulations on the cost of financing and the capital structure of banks, the literature has not thoroughly assessed whether this impact extends to regular corporations. A notable exception is the work of Gropp et al. (2019), who analyzed the effects of capital constraints in banks, resulting from regulations, on the corporations that are clients of these banks. They found that these firms exhibit lower asset, investment, and sales growth. The proposed project aims to contribute to the literature by investigating an unexplored aspect related to firms' capital and debt structures using a regression discontinuity approach (please refer to section 2.3 for detailed information). Furthermore, we will explore the impacts of Basel III finalization, a regulation that has not been previously studied in this specific context.

Mortgage lending standards and real estate market development

The second subproject “Mortgage lending standards and real estate market development” focuses on real estate markets and investigates the relation between leverage and real estate prices. Real estate markets have been documented to be plagued by frictions. The transparency is low (Schulte et al., 2005), transaction costs are high (Van Ommeren and Van Leuvensteijn, 2005) and markets are segmented (Beracha et al., 2018). The previous arguments together with the difficulty in shorting real estate assets implies that it takes longer for real estate prices to return to fundamentals. Due to this it is generally agreed that real estate markets are subject to bubbles (e.g., Brunnermeier and Julliard, 2008). This is worrisome for economists because it has been documented there is a close link between bubbles bursting and broader financial crises. A prime example of this is the US subprime crisis and the following great financial crisis (Brunnermeier, 2009).

It is often argued that lending standards and leverage play an important role in fueling the bubble. There is a correlation between leverage and real estate prices. However, it is not clear in which direction the causality runs. Gan (2007) argues that real estate assets serve as collateral for mortgages. Once the real estate prices increase the value of the collateral increases and thus the mortgage volume and leverage increase. On the contrary other researchers such as e.g., Pavlov and Wachter (2011) argue that causality runs the other way around. Their argument states that increased leverage leads to higher demand for real estate and thus higher prices. There are numerous studies that have investigated the relation between mortgage market conditions such as e.g., leverage restrictions and the housing

markets (see e.g., Poghosyan, 2020 for a literature review in table 1a) mostly finding that the mortgage market regulations measures can cool down markets. However, these studies cannot establish causality because they do not observe the counterfactual.

Quasi-natural experiments can help to establish causality. However, comparably few studies (see e.g., Poghosyan, 2020 for a literature review in table 1b) in the field use these identification mechanisms. Notable exceptions are Armstrong et al. (2019) who study the effect of loan-to-value ratio restrictions on house prices in New Zealand in a difference-in-difference framework and Deng and Gyourko (2019) who study several cooling measures in Singapore using a regression discontinuity design.

The proposed research project will study the introduction of a self-regulation by the Swiss banks that effectively limits the eligible equity that is used to calculate the maximum amount of leverage in certain real estate market segments. Using a difference-in-difference framework we can identify a causal relationship from leverage restrictions to real estate prices. We contribute to the literature in two ways. First, we investigate a rule change of a different nature compared to the existing literature (see section 2.3 for details). This enables us to uncover the relative importance of the different channels that drive the relationship between leverage and housing prices. Second, we think our identification approach is superior since the comparability of the market segments that serve as treatment and control group in the difference-in-difference is higher than in the previously studied markets, given the market structure in the Swiss real estate market. Moreover, we use a novel approach to control for the factor structure.

Apart from this, the subproject contributes to the literature by investigating the impact of financial regulation on leverage. This has barely been studied. A notable exception is Hoesli et al. (2020). The paper shows that public real estate returns react to changes in financial regulation using event study methodology. Our paper contributes to this strand of the literature by investigating private markets and shedding lights on the leverage channel that drives this reaction. Studying private markets is important since they are larger than public markets and private borrowers (households) are often highly leveraged and sensitive to interest increases. In addition to that, the results of this subproject will also yield insights regarding the questions whether financial regulation can negatively impact real estate markets when it comes at the wrong time (i.e., initiate the burst of a bubble) and whether it has adverse consequences for the distribution of wealth in the society. If lending standards are too tight people with little wealth are not able to afford real estate. This is particularly relevant for countries such as Switzerland that have very low homeownership rates and implemented measures with a special design.

Bank regulation and the maturity of loans

The third subproject is concerned with the maturity choice of borrowers. Naturally the paper is related to papers that investigate the determinants and implications of debt structure. Diamond (1991) investigates the relation between debt maturity and liquidity risk, Bolton and Scharfstein (1996) model the optimal number of borrowers as a tradeoff of restructuring costs and bargaining power, Diamond (1993) investigates the choice of debt seniority. Apart from this there are several theory papers that investigate the optimal debt capital structure (Park, 2000; Bolton and Freixas, 2000 and DeMarzo and Fishman, 2007). Rauh and Sufi (2010) show that changes in debt capital structure are underestimated

if debt heterogeneity is not considered. Recently Benmelech et al. (2022) have shown that a secured credit premium impacts issuance activity of secured debt.

Apart from the above-mentioned literature on the debt capital structure the paper is related to the strand of the literature that investigates the impact of capital requirements on banks (see also in the section Bank Regulation and corporate financing decisions). The proposed project adds to the literature by uniting the two strands of the literature. The project attempts to show that financial regulation can alter the choice of loan maturity a quantity that has not been studied in this context so far. The focus on financial regulation as a driver of debt capital structure is novel. Moreover, we add to the literature by investigating the issue based on bank portfolio data instead of corporate data. We argue that the choice of the maturity is important because it is associated with rollover risk and liquidity risk (Diamond, 1991 and Archarya et al., 2011). Moreover, from a methodological perspective the proposed use of the GIV approach (Gabaix and Koijen, 2020) is novel (see 2.3 for details).

2.2. Current state of own research

The proposed project is a collaboration between the University of Zurich (UZH) and the Zurich University of Applied Sciences (ZHAW). The completion of this research agenda requires both extensive knowledge of regulatory frameworks, insights into the Swiss real estate market as well as expertise in panel data econometrics.

Per Östberg has studied how to apply difference-in-differences when the data generating process has a factor structure. The results are published in the working paper "Difference-in-differences with Economic Factors and the Case of Housing Returns" (Huang and Östberg, 2023). The paper shows that the existing literature does not adequately consider the factor structure present in real estate returns and develops alternatives that are unbiased and examines the performance of these estimators. Apart from the above-mentioned working paper Per has a track record of publishing research papers that use large and complex panel and time series data sets as well as cutting edge econometric identification schemes (e.g., Östberg and Richter, 2018; Hvide and Östberg, 2015; Nyborg and Östberg, 2014 and Östberg and Bodnaruk, 2009) in leading academic journals. Per's expertise in combining difference-in-differences with factor models is necessary for the identification approaches in the project.

The second necessary component is the understanding of the modern regulatory framework and its implications for financial intermediaries and ultimately broader financial markets. Prior to joining ZHAW Thomas Richter worked at a leading Swiss bank where his job entailed AIRB model development (loss given default model for the Swiss mortgage portfolio and a probability of default model for income producing real estate mortgage segment).

Thomas current research agenda at ZHAW focusses on the impact of bank regulation on mortgage and real estate markets (Richter und Ziegler, 2021; Richter und Ziegler, 2022) and bubble risk in real estate markets (Müller et al., 2023). The focus of the research is on applied topics that are of relevance for banking and real estate professionals in Switzerland. During his doctoral studies at the UZH (supervised by Per Östberg) Thomas has investigated the effect the enactment of the Basel III leverage rule on the use of CCP and the market making activity in the European sovereign bond market (Richter, 2019 and Richter, 2021).

These papers and the interaction between UZH and ZHAW colleagues have shaped the agenda for the proposed project. While writing the above-mentioned papers three areas for future research have been discovered: 1) regulation has adverse effects on mortgage and real estate markets, 2) recent regulatory changes offer unique opportunities for the identification of causal effects and 3) the research results are of great interest for practitioners and policymakers.

2.3. Detailed research plan

Bank Regulation and corporate financing decisions

Economic mechanism and motivation

With the Basel III finalization coming into force in the beginning of 2023 the use of the AIRB approach is no longer allowed for large and mid-sized corporates with consolidated revenues that exceed EUR 500 million. The major change coming from this is that banks are no longer able to estimate their own values for the loss given default (LGD) parameter. The relevant Basel III paragraph states that senior claims on corporates that are not secured by recognized collateral will be assigned a 40% LGD (BCBS, 2017a).

Previous academic literature has demonstrated that banks have utilized their discretion in estimating their own risk parameters to reduce the required regulatory capital (Plosser and Santos, 2018). The regulator's stated objective is to eliminate this discretion through the finalization of Basel III. One of the measures employed for this purpose is the use of a fixed 40% LGD for unsecured (senior) corporate exposures (BCBS, 2017b). To assess the impact of these reforms, the BCBS is conducting a series of quantitative impact studies (QIS). The results of these studies indicate a substantial increase in capital requirements due to the Basel III finalization for large European and American banks (BCBS, 2023). The aspect of the regulation described above, concerning LGD, is likely to be a contributing factor to this increase. Furthermore, anecdotal evidence from the banking industry suggests that LGDs for corporates transitioning away from the AIRB approach are on the rise. Therefore, we hypothesize that it is probable that LGDs under the new regulatory regime will be higher on industry average.

As highlighted by a referee, our evidence regarding whether the modeled LGDs under the AIRB approach are higher or lower than the new ones provided by the regulator is currently incomplete. Further verification of our hypothesis will be carried out during the project. The data obtained from Pillar 3 disclosures, which are collected as part of sub-project 3 (as described below), will enable us to conduct a comprehensive assessment. From a project risk management perspective, it is essential to emphasize that while an increase in LGDs, as suggested e.g., by the results of Plosser and Santos (2018), is likely, it is not critical for our research projects. Our primary requirement is to identify a discontinuity at the EUR 500 million threshold.

Generally, we only need to identify a discontinuity, which is clearly present at the EUR 500 million threshold. Additionally, the referee noted that the change can vary from bank to bank, with some banks modeling AIRB LGDs very conservatively potentially facing lower LGDs, while others may need to increase their LGDs following the rule change. We agree with this assessment. However, our intention is to determine the industry-wide effect and base our research on this. If it proves difficult to establish

an industry-wide effect, we will identify impacted market segments, such as specific countries or country groups, and proceed accordingly. In such cases, the research project will remain feasible. Given the assumption of increasing LGD for former AIRB exposures we hypothesize that regulation will result in higher cost of capital for banks in this segment which in turn will result in higher prices for unsecured loans (secured loans are not or less impacted by the rule change). We therefore predict that companies with revenues of more than EUR 500 million (big companies) use significantly less bank financing compared to companies just below the threshold that are not subject to the new regulatory rules. Apart from that, it can be expected that the companies above the threshold use relatively more secured bank financing. In addition to that, we expect to see a tendency for short term bank financing in the transition period. A 1-year loan granted in 2020 is not impacted by the rule change but a 5-year loan granted in 2020 is, since it is still on the balance sheet when the rule was enacted in the beginning of 2023. In general, this rule change makes debt less attractive for borrowers that rely on bank loans. Consequently, we also expect to uncover decreasing leverage for the companies above the threshold.

In short, this subproject is set out to show that the recent regulatory changes influence corporate capital structure decisions. While it might make sense that banks have less discretion in determining their risk parameters and therefore capital requirements it is not clear whether the regulatory rules should be implemented in a way that alters the incentives to finance in the corporate sector. This might lead to destabilizing effects since it fosters disintermediation, a shortage of collateral and maturity mismatches between liability and asset side in corporate balance sheets. Moreover, it alters relative prices for companies with different economic but the same regulatory LGD.

Methodology

The proposed project will use a regression discontinuity approach to show that around the EUR 500 million threshold there is a significant difference in the (debt) capital structure of companies (discontinuity). Using the RD approach a causal relationship can be established if certain assumptions of the RD design are met (see e.g., Cattaneo and Titiunik, 2022). If the assumptions of RD hold, then all observed and unobserved covariates are balanced and therefore classical RD designs do not include covariates. We estimate the following:

$$y_i = \alpha + \gamma X_i + h(Z_i) + \epsilon_i.$$

In this specification y_i is the variable of interest. This could be the debt-to-equity, secured loans to total debt, share of long-term loans to total debt or share of bank loans to total debt. The variable X_i is an indicator variable that takes the value of one when the revenue is above EUR 500 million and zero otherwise. The variable Z_i are the revenues and $h()$ is a continuous function typically a higher order polynomial.

Robustness checks and mitigation of project risk mitigation

In response to several good referee comments after the first submission of the proposal we now plan a battery of robustness checks to ensure the validity of the projects results. The most important ones are sketched below:

Alternative methodology: To ensure that our results are robust to the use of different identification methodologies we will run a difference-in-difference regression where we use the firms with above EUR 500 Mio as the treated group and the one below as the control group. Many papers in finance use a combination of the RD and the DiD approach (e.g., Black and Kim, 2012). As explained above we also plan to identify segments (e.g., countries or industries) that are more impacted than others by the rule change. We will assess this in a triple difference-in-difference setting as in Muehlenbachs et al. (2015), which will further strengthen the support for our hypotheses.

Placebo tests: To ensure that the discontinuity is not driven by other variables such as e.g., a size effect (market capitalization) we look for discontinuities in other relevant variables that determine capital structure. Moreover, we run the RD for short term loans before the enactment of Basel III. We expect that the discontinuity is not visible there. Which would further strengthen our argument, that the rule change is valid for identification of our effects of interest.

Other regulatory changes: At this point we are certain that the identified rule change is the dominating one within the Basel III framework for the quantities in question. However, as part of the project we will also further investigate the institutional environment. In case we find other regulatory changes that might distort our results (project risk) we can use the difference in discontinuities approach as suggested by Grembi et al. (2016).

External validity: Low external validity is a limitation of the RD design. However, as summarized in Cattaneo and Titiunik (2022) there are approaches that enable valid extrapolation of RD treatment effects. For our purpose the extrapolation method via the derivative of the RD average treatment effect seems feasible.

Control variables: We will also estimate the RD regression with covariates. We follow the approach recommended in Calonico et al. (2019). The rich databases offer dozens of relevant control variables.

Data

The study will cover several thousand companies over several years. The data will come from databases such as CRSP, Capital IQ and Compustat (already available at the UZH). However, this dataset will be supplemented with information on outstanding debt products obtained from the Mergent Fixed Income Securities Database (see in the description of the budget why this database is important for the project). This additional database is essential for conducting a detailed analysis of the debt capital structure. We will have access to quarterly data. We anticipate that we will have good data coverage for publicly listed companies in the 27 jurisdictions that adhere to Basel regulations. The IRB status of a bank (FIRB or AIRB) can be ascertained from the CR6 table found in the bank's mandatory Basel Pillar 3 disclosures.

To estimate the RD equation stated above, we need a cross section of banks after the regulation was initiated. For the described robustness tests, we even require a panel data set. By the start of the project, we will have a panel data set of at least five quarters before and after the rule change available in our databases. This should be sufficient to identify the effect even though bank responses might be slightly delayed. Moreover, the rule change was announced a long time before and banks had time to prepare.

Also, during the research project, we can add additional data points to further rule out that we miss delayed responses. It should also be noted that banks needed to account for the rule changes earlier because banks knew that long term loans granted before the rule change will be on the loan books until after the regulation will be enacted.

Project team, roles, and outcomes

The research team for this sub-project will consist of Per Östberg, Thomas Richter, the UZH-based PhD student, and the postdoctoral researcher. Overall, we anticipate that this subproject will yield three papers along with several conference presentations, seminars, and workshops. The first paper will serve as the flagship publication of this subproject and will follow the methodology and content described above. Our goal for this paper is to present it at leading academic conferences and ultimately publish it in a top-tier finance journal. Given the complexity of the regulatory framework, the size and number of the considered databases, and the associated econometric challenges, this paper will be the most labor-intensive within the subproject and will therefore involve the entire sub-project team.

Per Östberg and Thomas Richter will share responsibility for the research design and the writing process. Per Östberg will lead in determining the appropriate econometric specifications and communicating the results to the academic community. Thomas Richter will ensure a strong link between the econometric models and the institutional aspects of the regulatory framework, as well as communicate the results to practitioners. Additionally, both researchers have a strong understanding of the existing literature in this field and can contextualize the proposed research within it, clearly highlighting its contribution. The postdoctoral researcher will handle the estimation of the econometric models and will also serve as a valuable sparring partner for refining the research design and the paper's storyline. The PhD student will be responsible for data collection and preparation and will also contribute to small work packages in other areas such as estimation, literature review, and writing, which will aid in their development. Given the substantial contributions of the postdoctoral researcher and the PhD student, both will be recognized as co-authors of the paper (see 2.4 for milestones and detailed planning).

In addition to the aforementioned research paper, the topic of "Bank lending and corporate financing decisions" is underexplored, and the research area we have outlined offers ample opportunities to investigate several closely related questions. The PhD student hired for this project will have academic freedom to write 2 additional academic papers that meet the standards of the UZH Department of Banking and Finance for granting a PhD degree. They will receive supervision from Per Östberg and benefit from the advice of Thomas Richter. PhD papers at UZH often result in journal publications and PhD students interested in an academic career can reach academic placements.

Mortgage lending standards and real estate market development

Economic mechanism and motivation

It is often hypothesized that increasing levels of debt financing led to real estate price bubbles (e.g., Mian and Sufi, 2011). However, other strands of the literature argue that increased real estate prices mechanically lead to an increase in debt because the value of the collateral changes (e.g., Gan, 2007 or Shleifer and Vishny, 1992). In other words, there are arguments in favor of causality in both directions.

The Swiss mortgage market is a good laboratory to study the impact of debt financing on mortgage prices. A change in the self-regulation in Switzerland offers a unique opportunity to identify the effect of changes in leverage on housing prices. The lending standards of the Swiss banks for mortgages require borrowers to bring a certain equity stake. Currently borrowers need to have at least 20% equity when borrowing for self-occupied real estate. Until 2012 the banks potentially allowed that all required equity was sourced from pension fund accounts. However, in 2012 the regulation was changed, and the banks required that borrowers bring at least 10% equity that is not sourced from pension fund accounts. This regulation only affected the self-occupied real estate segment.

An industry study estimated that approximately one-third of real estate purchases in Switzerland are funded using pension funds (MoneyPark, 2017). Analyzing the FINMA (Swiss Financial Market Supervisory Authority) communication regarding the rule change shows that it was implemented predominantly to regulate risk-taking in the banking sector and not influence the real estate sector directly (FINMA, 2012). The project deliveries will also include an in-depth analysis of the institutional environment to assess the validity of the rule change in the foreseen methodological framework.

This change in the self-regulation can be viewed as newly introduced constraint to leverage. We exploit the fact that the new regulation only impacts the condominiums in self-occupied real estate segment (SORE)² but not the multi-family home income-producing real estate sector (IPRE). We argue that multi-family homes and condominium prices should behave in the same way in the absence of market frictions. This argument is natural since a multi-family home consists of several condominiums (law of one price).

The relationship between leverage and housing prices has been extensively studied in previous literature (see section 2.1 for the current state of the literature). Our contribution to this body of research lies in our exploration of the unique characteristics of the Swiss housing market and its rules and will enable us to generate additional insights of general relevance for the understanding of the underlying economic mechanisms. Most previous studies focus on events where the maximum loan-to-value ratio or taxes are altered, typically resulting in negative impacts on the housing market. These impacts are driven by three key mechanisms: 1) reduced affordability, 2) changes in the rent-or-buy decision, and 3) diminished attractiveness of housing investment relative to other opportunities, particularly when a high leverage ratio is considered optimal. Unlike previous studies, we can attribute the changes to specific mechanisms due to the nature of the rule change we have described (change in the definition of eligible equity). In our case, only the definition of assets has changed, not the maximum leverage (loan-to-value ratio). Consequently, we can rule out mechanisms 2) and 3) as drivers of the results, shedding light on the relative importance of the underlying mechanisms that seem to influence outcomes in the existing literature.

Furthermore, we can examine how this rule change affects different population groups and regions. We hypothesize that lower-income individuals are more affected because they have fewer alternative assets

² It also impacts the single-family homes. However, since the condominiums are more comparable with the multi-family homes that are not impacted, we focus on them.

to substitute for funds in their pension accounts. As a result, we anticipate that the luxury housing market will be less affected. Demonstrating this will contribute to the literature by illustrating how the design of a mechanism can influence its impact on specific population groups, potentially implying welfare effects. Additionally, as noted by a referee in the initial review, regions with lower housing prices are less impacted mechanically. We will also explore how this change might induce regional divergence in housing prices.

In addition to these contributions, we believe that the Swiss housing market offers advantages for identifying the effects of such a rule change. Many previous studies do not use quasi-experimental approaches to identify the effect, raising questions about the accuracy of their findings in establishing the relationship between leverage constraints and housing prices. While some studies employ quasi-experimental approaches, as highlighted by Deng et al. (2019), it can be challenging to cleanly identify treatment and control groups. Switzerland's housing market, with its developed rental market in the condominium segment and a high proportion of privately owned apartments, provides two market segments that are highly comparable.

Although the change we investigate is specific to the Swiss housing market, the results of our study are of broader interest beyond Switzerland. They can offer general insights into the relationship between housing prices, bank regulation, and leverage. These factors have received significant attention in the literature thus far, and our research can identify the mechanisms driving the results of previous studies and shed light on the importance of the design of cooling measures for welfare purposes.

Methodology

To detect the effect of leverage on real estate prices in Switzerland we use the following difference-in-difference regression:

$$Return_{i,t} = \alpha + \beta_1 \times Post_t + \beta_2 \times Treat_i + \beta_3 \times Treat_i \times Post_t + \sum_{k=1}^K \gamma_{i,k} \times Factor_{k,t} + \epsilon_{i,t}$$

The $Return_{i,t}$ is the return in MS region i at time t . The variable $Post_t$ is a dummy variable that takes the value of 1 when the observed house price dates from before the rule change (2012) and 0 otherwise. The variable $Treat_i$ is a dummy that takes the value of 1 if the observed price i belongs to the SORE condominium segment (treated group) and 0 otherwise (control group). The $\beta_3 \times Treat_i \times Post_t$ denotes the interaction of the two variables. This means β_1 captures different return developments in the pre- and post-regulation period (can be replaced by time fixed effects), β_2 captures the different in the price level across the segments (can be replaced by property type fixed effects and/or region fixed effects), β_3 identifies the different price development in the two segments. If the difference-in-difference assumptions are satisfied β_3 identifies the treatment effect. Following Huang and Östberg (2023) we optimally select and control for K factors that explain housing returns and estimate loadings for each region.³ Moreover, we control for covariates that explain the differences in prices developments (apart from the treatment). Most notably the databases include variables with information about the real estate

³ We use MS (mobilité spatiale) regions as defined by the Swiss federal statistical office. There are 106 MS regions, and they are used as an intermediate microregional level for numerous scientific and regional policy purposes.

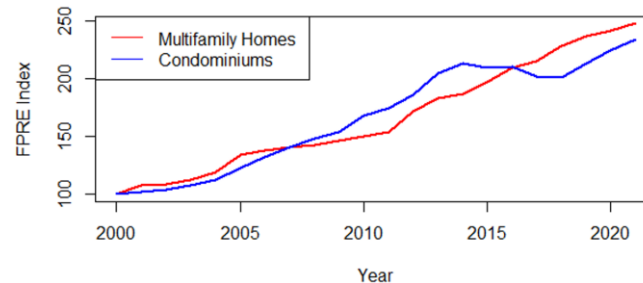
type (e.g., multi-family-home), object data (e.g., building year and square meters), information about the location and transaction details (price).

Robustness checks and mitigation of project risk mitigation.

In robustness checks, we will verify the satisfaction of the parallel trends assumption in the difference-in-difference approach. Jointly with the investigation of the institutional environment this will verify the validity of our methodology. Additionally, we will cluster the standard errors at appropriate levels and collapse the time dimensions as suggested by Bertrand et al. (2004). Furthermore, we will utilize triple interactions to demonstrate that the effects are more pronounced in segments that are relatively more affected by the rule change, such as low-priced homes where borrowers are likely to be more constrained by leverage and low-price regions where minimum equity requirements are less relevant compared to high-price regions.

As noted by a referee in the first round, the consideration of the factor structure in the difference-in-difference approach, as proposed in Huang and Östberg (2023), has not yet been published in a peer-reviewed journal. To ensure that the project results are accepted by the research community, we will revert to the classical difference-in-difference setup if the proposed methodology is not accepted by peers. If the publication is accepted, we will have an additional methodological contribution by applying it in a relevant context.

Figure 1: Development of real estate prices in the multifamily home and condominium segment



Data

The study will be based on transaction level data from the Swiss Real Estate Data Pool (SRED) and the Real Estate Investment Data Association (REIDA). The SRED dataset contains approximately 2000 transactions per quarter and goes back until 2002. Besides the transaction prices the dataset contains more than 40 variables. The REIDA Dataset will be used to enrich the SRED dataset because it has better coverage of multifamily homes. This dataset covers to 2011-2023 period (the databases and the variables contained in it are further described in the budget).⁴

Illustration of the effect

Figure 1 shows the development of the FPRE Index for multifamily homes and condominiums. It can be clearly seen that they move in parallel until 2012. However, once the leverage constraints hit the SORE

⁴ The coverage of MFH transaction in the REIDA data set is not entirely clear. In case this data does not contain enough transaction data an alternative data set with index data on MS region level since 2000 (yearly) or 2010 (quarterly) can be used alternatively (Source: Fahrländer und Partner Raumentwicklung (FPRE). Moreover, we explored the option of using offering prices for our study and we think also this approach will be feasible. This data can e.g., be purchased from Immolex.

segment the price development diverges. This very aggregate graphs suggests that the difference-in-difference approach that is supposed to be applied in the project is valid (no other visible material events and parallel trends in absence of treatment).

Project team, roles, and outcomes

The research team for this sub-project will consist of Per Östberg, Thomas Richter, and the ZHAW-based PhD student. The division of work will be similar to that outlined for subproject 2. The primary paper resulting from the project is expected to meet the quality standards for publication in a high-ranking real estate or finance journal. Per Östberg and Thomas Richter will be responsible for the research design, writing, and literature review. Per Östberg's methodological knowledge is required for the difference-in-difference analysis as proposed by Huang and Östberg (2023). Additionally, Thomas Richter's expertise in the specificities of the Swiss real estate and mortgage market is essential.

Like sub-project 1, the PhD student will handle data collection, data preparation, model estimation, and contribute to various smaller work packages in areas such as estimation, literature review, and writing, which will contribute to their professional development. The rich dataset acquired for this sub-project and the broader topic of "Mortgage lending standards and real estate market development" offer the potential for two additional papers that will be included in the PhD thesis of the ZHAW-based PhD student as part of their involvement in the project. The PhD student will collaborate with Thomas Richter at the ZHAW School of Management and Law and choose a university professor from a Swiss university to supervise their dissertation. Per Östberg at UZH will be available for supervision and advice. The research paper quality will be ensured by the degree-granting institution, guaranteeing publishable standards (for the projects timeline and milestones please revert to section 2.4).

The ZHAW-based PhD student will develop a more applied profile. To support this, they will collaborate with Thomas Richter to write articles for practitioner journals, newspapers, and a white paper summarizing the key findings of the academic project for regulators and practitioners. Additionally, they will organize a science-to-practice event or conference where the research project outcomes will be discussed among academics and practitioners.

Bank regulation and the maturity of loans

Economic mechanism and motivation

The third subproject focuses on bank regulation and the maturity of loans. The current Basel regulation incentivises corporations and AIRB banks to agree on short-term loans. The bank uses a formula given by the regulator to determine the capital requirements arising from loans. This formula contains a so-called maturity adjustment. Due to this adjustment the capital requirements for the bank increase with the maturity of a loan. This in turn leads to an increase in the price for long-term loans due to cost of capital considerations of the bank when pricing a loan. This makes sense because long-term loans are riskier. From an economic perspective, some loans such as mortgages are long-term loans. However, from a legal perspective it is possible to replace a long-term loan by a sequence of short-term loans. The contractual (and not economic) maturity is relevant for the calculation of the maturity adjustment.

To avoid paying the high cost of capital corporate clients often prefer to choose a sequence of loans with shorter maturities as opposed to a long-term mortgage. Potentially, this choice makes them more

vulnerable to interest rate and refinancing risks. However, the choice of a shorter (legal) maturity of loan contracts does not reduce the banks risk because in a downturn state of the market the borrower cannot refinance elsewhere, repay, or liquidate the collateral that secures the loan. Because the maturity adjustment increases disproportionately with the probability of default borrowers with low credit quality are affected more.

The internal ratings-based approach (IRB) consists of an advanced approach (AIRB) and a foundation approach (FIRB). Banks that used the AIRB approach use the actual legal maturity of a loan. Banks that use the FIRB approach must use a fixed maturity of 2.5 years. This means that for some banks the maturity matters and for others not. This may result in different maturity structures of the loan portfolios of FIRB and AIRB banks.

It should be noted that banks face significant hurdles when they wish to switch their approach, such as moving from AIRB to SA or FIRB. Any changes must be approved by local supervisors. For European banks, this requirement is outlined in Article 149 of the Capital Requirements Directive (CRR), which states that “. An institution that uses the IRB Approach for a particular exposure class or type of exposure shall not stop using that approach” A switch is only possible if certain conditions are met. Among other requirements, the bank needs to demonstrate that they are not switching to reduce capital requirements (Deloitte, 2023). This significantly limits selection issues in our research design.

Methodology

We exploit these particularities of the regulatory framework in a baseline fixed effect regression where we explain the loan-maturity m of a loan portfolio p of bank i on time t with a group dummy D and control variables.

$$m_{p,i,t} = \alpha_i + \alpha_p + \alpha_t + \beta D_i + \sum_{k=1}^K \delta_k \text{Control}_{i,p,t} + \epsilon_{i,p,t}$$

The group is determined by the IRB status of the bank (FIRB or AIRB). The idea is to show that loan portfolios that are subject to a maturity adjustment have longer maturities than those that are not (but are otherwise identical). The potential control variables on portfolio level p include the PD, LGD, Credit Conversion Factors, the average Loan Size and exposure data.

The classical fixed effects approach has the disadvantage that we cannot estimate supply and demand elasticities since we only observe the equilibrium outcome. Therefore, we also propose to analyze the quantities of interest in a granular instrumental variable (GIV) framework, as exemplified by Galaasen et al. (2020), Chodorow-Reich et al. (2021), or Camanho et al. (2022) in other contexts. Generally, the GIV approach disentangles systematic and idiosyncratic components of a variable. The idiosyncratic part can serve as an "instrument" to estimate causal elasticities. We first apply the approach to a time series of bank capital ratios to measure idiosyncratic shocks to the bank's capital, following the methodological approach outlined in Gabaix and Koijen (2020). Next, we will replace the variable "D" in the regression above with a variable derived from these shocks, ultimately allowing us to estimate the effect of bank capital scarcity on loan maturity. By design of the GIV, the constructed variable measures bank-induced supply effects.

Robustness checks and mitigation of project risk mitigation

In robustness checks, we aim to show that (1) the effect increases with the probability of default (because the maturity adjustment increases with the PD) and (2) that capital constraint banks are more likely to have lower maturity loans than well capitalized banks. These effects can be identified using interactions of the D variable with the PD or a measure for capital constraints of the Banks such as RWA divided by Tier 1 capital.⁵

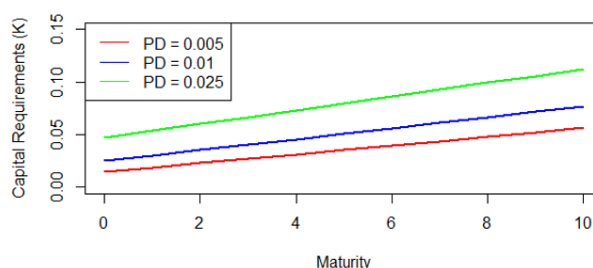
Data

The CR6 table of the Basel Pillar III reports contain standardized data about exposures, maturities, and risk parameters (such as e.g., loss-given default) on loan portfolio level. We will use an international sample of IRB banks in jurisdictions that follow Basel 3. We expect good data coverage for at least 179 banks that participated in the BIS Quantitative Impact Study (BCBS, 2023). The CR6 table is published since 2016 on a semi-annual basis (BCBS, 2018). Each report contains information on different segments and portfolios. The estimated sample size is several ten thousand of observations. We will collect these data using web-crawling technology from the bank's websites, where they publish their pillar 3 reports. This data set can be made publicly available and support the research of others. Moreover, we will enrich this data set with several WRDS databases (already available at the University of Zurich). In the Budget we further elaborate on how we want to make this data public.

Illustration of the Effect

The capital requirements K are calculated based on the LGD, the PD, the asset correlations R , and the maturity M . The Figure below plots the capital requirements for different PDs and maturities. The capital requirements increase with maturities (movement on the curves) and with the PD.⁶

Figure 2: Loan maturity, capital requirements and probability of default



Project team, roles, and outcomes

This sub-project is led by the postdoctoral researcher, who will oversee data collection, research design, and estimation. Thomas Richter and Per Östberg will provide support to the postdoctoral researcher in these tasks. Furthermore, they will play an active role in writing and shaping the contribution, as they

⁵ Another way to identify the effects from the maturity adjustment would be to compare the maturity structure of segments that are subject to the adjustment to those that are not but otherwise very similar. An example is for example private mortgage vs. corporate mortgages.

⁶ The capital requirements are calculated with $K = \left[LGD \times N \left[\frac{G(PD)}{\sqrt{(1-R)}} + \sqrt{\frac{R}{1-R}} \times G(0.999) \right] - PD \times \right. \\ \left. LGD \right] \times \frac{(1+(M-2.5) \times b)}{(1-1.5 \times b)}$ with $b = [0.118552 - 0.0548 \times \ln(PD)]^2$, the LGD and R are fixed.

originally conceived the idea. Both PhD students will also contribute to the project. All five participants in the project will be listed as co-authors.

This project requires the support of a postdoctoral researcher with expertise in artificial intelligence and web crawling capabilities, as the data acquisition process is complex and labor-intensive. Initially, we need to identify all the IRB banks and locate their Pillar 3 reports. This primary responsibility falls upon the two PhD students. Subsequently, the postdoc will develop a web crawler to download the Pillar 3 disclosure reports for the identified locations. With the assistance of artificial intelligence (AI), the postdoctoral researcher will then write code to extract portfolio-level information from the CR6 reports. While the reports have some degree of standardization, the tables may not always be identical in terms of presentation. The use of AI will minimize the need for manual intervention by the project team and streamline the process of updating the database with new reports. Ultimately, this will result in a database with tens of thousands of observations, including the variable of interest, information about the IRB status and a battery of control variables at the portfolio level (see section 2.4 for the detailed timeline of the project and the most important milestones).

The estimation of the baseline regression will be carried out by Thomas Richter and Per Östberg. The postdoctoral researcher will handle the GIV estimation. Given the novelty of the GIV estimation technology, there is no readily available implementation in standard statistical software. Additionally, this research project appears to be the first to apply this technology in the subfield of financial stability and regulation. To fully leverage the novelty of this approach, rapid implementation is crucial. Therefore, having a postdoctoral researcher with expertise in quantitative finance is essential for the project, as they are much faster than PhD students or the applicants (given their other responsibilities). Once the estimation is complete, the main writing tasks will be undertaken by Per Östberg and Thomas Richter. The postdoctoral researcher will provide important input, and the PhD students will assist with editing and refinement. Per Östberg will usually represent the research team at conferences.

Our goal is to publish the flagship paper of this project in a leading finance, management, or economics journal (FT50 list). Furthermore, we will make the database publicly available for use by other researchers. Thomas Richter will oversee this, in collaboration with the ZHAW PhD student. Additionally, the postdoc will have the academic freedom and time to write an additional paper using this data within the project's framework.

2.4. Schedule and Milestones

The table chart below gives an overview about the most important milestones of the project or the sub-projects respectively and the responsible project staff. Note the following abbreviations for the project members: PO = Per Östberg, TR = Thomas Richter, PostDoc = UZH postdoctoral researcher, PhD 1 = UZH PhD student and PhD 2 = ZHAW PhD student. Please consider the chart below jointly with the description of the role of the members in 2.3 and the budget in the administrative part.

Research Plan

Task	Lead	Q324	Q424	Q125	Q225	Q325	Q425	Q126	Q226	Q326	Q426	Q127	Q227	Q327	Q427	Q128	Q228
Overarching Milestones																	
Data acquisition	PO & TR																
Start date project staff	PO & TR																
Research Institutional Environment	TR & PhD 2																
Independent PhD project	PhD1 & 2																
Independent PostDoc Project	Post Doc																
Graduation PhDs	PhD1 & 2																
Bank Regulation and corporate financing decisions																	
Literature Review	All																
Data Preparation	PhD 1																
Refinement research design	PO & TR																
Model estimation	PostDoc & PhD1																
Writing	PO, TR & PhD1																
Working Paper on SSRN	All																
Conference submissions	PO																
Polishing	PO, TR & PhD 1																
Journal Submission	All																
Mortgage lending standards and real estate market development																	
Literature Review	All																
Data Preparation	PhD 2																
Refinement research design	PO & TR																
Model estimation	PO & TR & PhD2																
Writing	PO, TR & PhD2																
Working Paper on SSRN	All																
Conference submissions	TR																
Polishing	PO & TR & PhD2																
Journal Submission	All																
Bank regulation and the maturity of loans																	
Identification IRB Banks	PhD1 & 2																
Literature Review	All																
Refinement Research Design	Postdoc PO, TR																
Development code web-crawling	PostDoc																
Data collection	PostDoc																
Data Preparation	PhD 1 & 2																
Model estimation	PO, TR, PostDoc																
Writing	PO, TR, PostDoc																
Working Paper on SSRN	All																
Conference submissions	PO																
Polishing	PO & TR																
Journal Submission	All																

2.5. Relevance and Impact

Finance researchers usually agree that externalities and asymmetric information impair the functioning of the financial sector (e.g., Titman, 2013). Moreover, at this point there is a broad consensus among academics, policymakers, and the public that there are signs of market failures (imperfections) in the financial system (e.g., Greenwald and Stiglitz, 1993). The 2007 financial crisis serves as a prime example. As a result, there is some consensus that financial regulation is useful. It is undoubted that the recent advances in financial regulation have increased the capital base of banks and therefore increased financial stability (e.g., Gauthier et al., 2012). However, identifying a market failure and the need for regulation is only the starting point.

Once the market failure has been discovered the challenge for regulators becomes to define the rules in such a way that they are cheap to implement and have no (severe) adverse consequences. History has shown that often unintended consequences only become apparent when the regulation is in place. It is therefore crucial for academics to constantly review and evaluate the regulation in place. Given the pace, the complexity and innovation of new regulatory rules there is a lot of need for research.

The previous literature was mainly concerned with the effects of regulation on banks itself. The main question of this strand of the literature is whether the regulation incentivises the risk taking of banks and financial institutions in a desired way. Most notably there is an impressive literature showing how non-risk-based capital regulation have ample undesired effects (e.g., Duffie, 2018 and Richter, 2021). The proposed projects are distinct from this literature by focusing on the effects of banking regulation on corporate financing and real estate markets.

The corporate finance literature so far has mainly studied the availability of financing in response to regulatory reforms. Most of the papers have concluded that there is no credit crunch (e.g, Peek and Rosengren, 1995). We acknowledge that despite the increasing number of regulation financing is still available. However, the project aims to show that it may only be available in different forms and/or that relative prices of different financing options are altered. For example, this may imply short-term loans instead of long-term loans (project 3), debt instead of equity or a collateralized loan instead of an unsecured loan (project 1). This means we contribute to the literature by studying important quantities that remain unexplored so far such as the maturity or the debt capital structure.

Real estate market developments have often played an important part in the emergence of banking crisis. A prominent example is the 2007 financial crisis (Mian and Sufi, 2011). Despite their importance our understanding of the relation between leverage, bank regulation and house prices is incomplete though. Using a unique rule change in Switzerland we shed led on the underlying mechanisms that govern the relation. The implications of the research project are therefore relevant beyond Switzerland.

Overall, the resulting research will yield important information for government institutions and regulatory authorities to make informed policy decisions. In a broader context it also has implications for households as understanding the interrelationship between leverage and house prices could lead to better designed housing policies.

The project results will be disseminated in leading scientific journals (3 papers for tier 1 journals and up to 5 papers of publishable quality by junior researchers) and academic conferences. Additionally, the results will be spread to stakeholder groups in the economy via white paper(s), events and/or webinars summarizing the main takeaways. Relevant content will furthermore be integrated into university teaching (master, bachelor, and executive education programs). Moreover, the results will be published via social media and articles (magazines & newspapers). A special edition of the ZHAW Real estate circle (a bi-yearly science to practice exchange forum) will also be used to make the results known among practitioners and the public.